

LAB_02 : Commands in Fedora for File and Folders

Q.1 Change permissions for file “exam1” that should look like the following:

The user (owner) should have all the permissions.

Members of the group should have read and write permissions.

Others should have read permission.

Write the command that you will use.

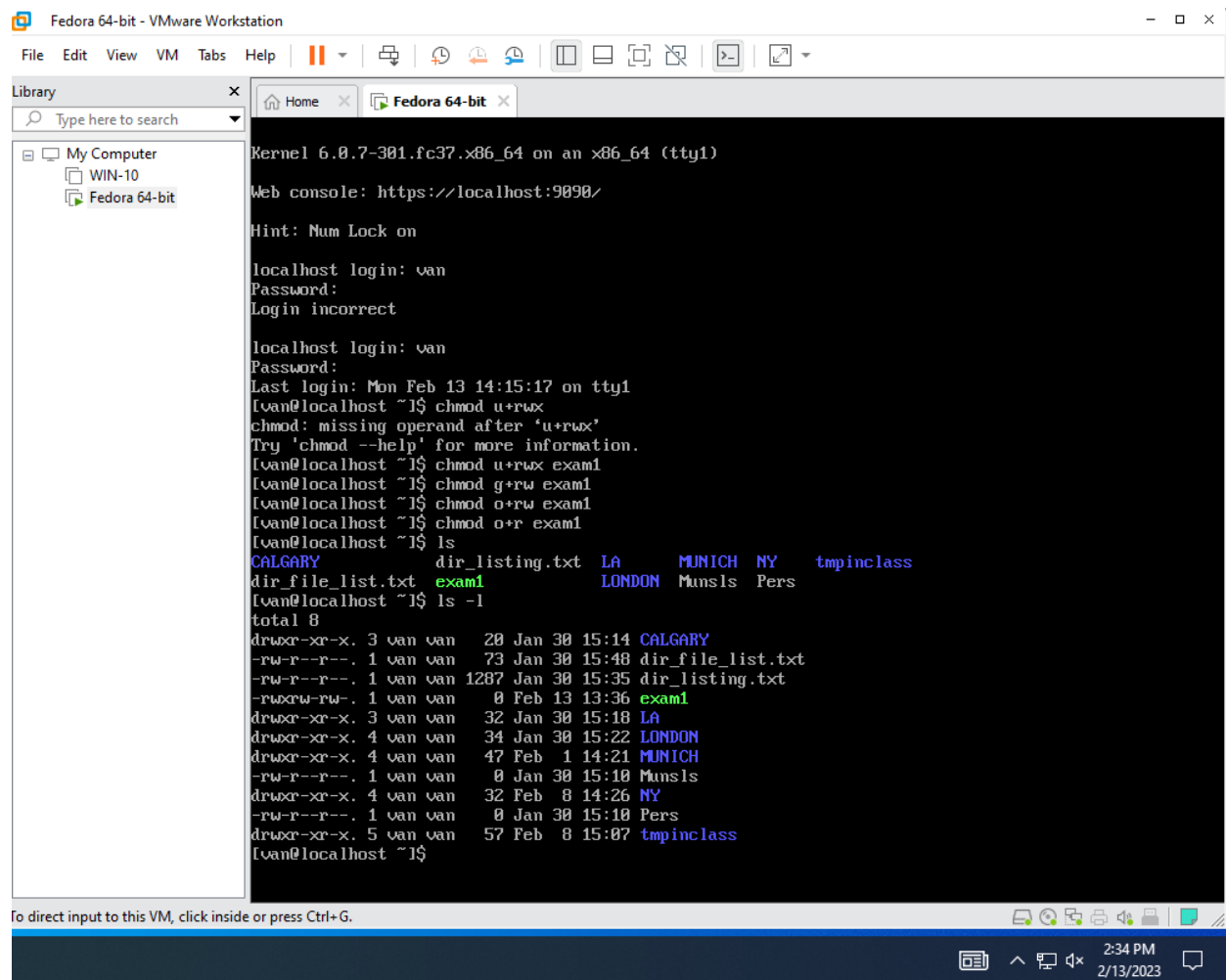
Answers Q.1 :

1. [`chmod u+rw` <file name>] : to add permissions for user own.
2. [`chmod -rw` <directory name>] : to remove permissions.
3. [`chmod +x` <filename>] : to allow executable permissions.
4. [`chmod -wx` <filename>] to take out write and executable permissions.

Note: that “r” is for read, “w” is for write, and “x” is for execute.

This only changes the permissions for the owner of the file.

```
0 --- no permission
1 --x executes
2 -w- write
3 -wx write and execute
4 r-- read.
5 r-x read and execute
6 rw- read and write.
7 rwx read, write and execute.
```



The screenshot shows a VMware Workstation window titled "Fedora 64-bit - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a Library pane on the left. The Library pane shows "My Computer" with sub-items "WIN-10" and "Fedora 64-bit". The main window displays a terminal session for the "Fedora 64-bit" VM. The terminal output is as follows:

```
Kernel 6.0.7-301.fc37.x86_64 on an x86_64 (tty1)
Web console: https://localhost:9090/
Hint: Num Lock on

localhost login: van
Password:
Login incorrect

localhost login: van
Password:
Last login: Mon Feb 13 14:15:17 on tty1
[van@localhost ~]$ chmod u+rx
chmod: missing operand after 'u+rx'.
Try 'chmod --help' for more information.
[van@localhost ~]$ chmod u+rx exam1
[van@localhost ~]$ chmod g+rw exam1
[van@localhost ~]$ chmod o+rw exam1
[van@localhost ~]$ chmod o+r exam1
[van@localhost ~]$ ls
CALGARY  dir_listing.txt  LA      MUNICH  NY      tmpinclass
dir_file_list.txt  exam1          LONDON  MunsIs  Pers
[van@localhost ~]$ ls -l
total 8
drwxr-xr-x. 3 van van 20 Jan 30 15:14 CALGARY
-rw-r--r--. 1 van van 73 Jan 30 15:48 dir_file_list.txt
-rw-r--r--. 1 van van 1287 Jan 30 15:35 dir_listing.txt
-rwxrw-rw-. 1 van van 0 Feb 13 13:36 exam1
drwxr-xr-x. 3 van van 32 Jan 30 15:18 LA
drwxr-xr-x. 4 van van 34 Jan 30 15:22 LONDON
drwxr-xr-x. 4 van van 47 Feb 1 14:21 MUNICH
-rw-r--r--. 1 van van 0 Jan 30 15:10 MunsIs
drwxr-xr-x. 4 van van 32 Feb 8 14:26 NY
-rw-r--r--. 1 van van 0 Jan 30 15:10 Pers
drwxr-xr-x. 5 van van 57 Feb 8 15:07 tmpinclass
[van@localhost ~]$
```

At the bottom of the window, there is a status bar with the text "To direct input to this VM, click inside or press Ctrl+G." and a system tray showing the time "2:34 PM 2/13/2023".



Q.2 What are the permissions that the user, group, and others have given the following command: `chmod 654 filename`.

Answer Q2 :

Chmod 654 (`chmod a + rwx, u-x, g-w, o-wx`) sets permissions so that,

(U)ser / owner can read, can write and can't execute.

(G)roup can read, can't write and can execute.

(O)thers can read, can't write and can't execute.

For files

After changing a file's mode to 654 the file's mode will be displayed in Unix style file listing as:

```
-rw-r-xr-
```

For folders

After changing a directory's mode to 654 the folder's mode will be displayed in Unix style file listing as:

```
drw-r-xr-
```

Now we want to change the permissions for this file to say, `rw--r-xr--`.

Owner permissions(`rw-`) = $4 + 2 + 0 = 6$

Group permissions(`r-x`) = $4 + 0 + 1 = 5$

Other user's permissions(`r--`) = $4 + 0 + 0 = 4$

[`ls -l <file name>`] : Check file name .

[`chmod 654 <file name>`] : To change the permissions of a file using

[`chmod 654 chmod.txt`] : changing the file permissions.

[`chmod <user type(u/g/o/a)> <add/revoke(+/-)> <permission type(r/w/x)>`

[**chmod** **u+rw** <chmod.txt>] To change the owner permissions of a file to read and write.

[**chmod** **a+w** <chmod.txt>] : To give a write permission to everyone

[**chmod** **o-w** <chmod.txt>] : To remove the write permission for others.

[**chmod** <permission> <directory name>] : To change the permissions of a directory.

[**chmod -R** <permission> <directory name>] : To change the permissions of a directory with its files and sub-directories recursively,

```

Fedora 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
WIN-10
Fedora 64-bit

ivan@localhost ~]$ chmod g+rw exam1
ivan@localhost ~]$ chmod o+rw exam1
ivan@localhost ~]$ chmod o+r exam1
ivan@localhost ~]$ ls
CALGARY  dir_listing.txt  LA  MUNICH  NY  tmpinclass
dir_file_list.txt  exam1  LONDON  MunsIs  Pers
ivan@localhost ~]$ ls -l
total 8
drwxr-xr-x. 3 van van 20 Jan 30 15:14 CALGARY
-rw-r--r--. 1 van van 73 Jan 30 15:48 dir_file_list.txt
-rw-r--r--. 1 van van 1287 Jan 30 15:35 dir_listing.txt
-rwxr-xr-x. 1 van van 0 Feb 13 13:36 exam1
drwxr-xr-x. 3 van van 32 Jan 30 15:18 LA
drwxr-xr-x. 4 van van 34 Jan 30 15:22 LONDON
drwxr-xr-x. 4 van van 47 Feb 1 14:21 MUNICH
-rw-r--r--. 1 van van 0 Jan 30 15:10 MunsIs
drwxr-xr-x. 4 van van 32 Feb 8 14:26 NY
-rw-r--r--. 1 van van 0 Jan 30 15:10 Pers
drwxr-xr-x. 5 van van 57 Feb 8 15:07 tmpinclass
ivan@localhost ~]$ chmod 654
chmod: missing operand after '654'
Try 'chmod --help' for more information.
ivan@localhost ~]$ chmod 654 exam1
ivan@localhost ~]$ ls -l
total 8
drwxr-xr-x. 3 van van 20 Jan 30 15:14 CALGARY
-rw-r--r--. 1 van van 73 Jan 30 15:48 dir_file_list.txt
-rw-r--r--. 1 van van 1287 Jan 30 15:35 dir_listing.txt
-rwxr-xr-x. 1 van van 0 Feb 13 13:36 exam1
drwxr-xr-x. 3 van van 32 Jan 30 15:18 LA
drwxr-xr-x. 4 van van 34 Jan 30 15:22 LONDON
drwxr-xr-x. 4 van van 47 Feb 1 14:21 MUNICH
-rw-r--r--. 1 van van 0 Jan 30 15:10 MunsIs
drwxr-xr-x. 4 van van 32 Feb 8 14:26 NY
-rw-r--r--. 1 van van 0 Jan 30 15:10 Pers
drwxr-xr-x. 5 van van 57 Feb 8 15:07 tmpinclass
ivan@localhost ~]$

```

To direct input to this VM, click inside or press Ctrl+G.

2:42 PM
2/13/2023

